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Sentiment analysis of movie reviews based on quantum convolutional neural networks

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Abstract

Sentiment analysis has gained significant attention in the research domain of language understanding. It offers valuable insights into public opinion, customer feedback, and user experiences, but the quantum machine learning (QML) approaches used in it are still lacked and in the theoretical stages. In this paper, we focus on evaluating one of the QML approaches, which is QCNNs in movie reviews classification; this model represents a quantum neural network design inspired by Convolutional Neural Network (CNN) and exclusively employing parameterized quantum circuits. However, classical text data need to be converted into quantum states; for this, we use three quantum encoding techniques, which are angle encoding, amplitude encoding, and instantaneous quantum polynomial encoding. We investigate how different QCNN models distinguished by the number of qubits used and the structures of parameterized quantum circuits can improve the performance of sentiment analysis classification across both artificial and real datasets of movie reviews.

Keywords Quantum convolutional neural networks · Quantum machine learning · Natural language processing · Sentiment analysis

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1 Introduction

Sentiment analysis is a process used in text categorization to analyze and extract information related to individual opinions and emotions and determine the information's polarity, such as positive or negative. It attempts to measure people's feelings expressed in written text, and it is used in various fields such as business, social sciences, and social media analysis, see [1–5]. Machine learning algorithms have been used in sentiment analysis and have shown varying performance depending on the type of data being analyzed such as K-Means, Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), Naive Bayes, SVM, RNNs, CNNs, and LSTM in different studies [6, 7]. CNNs have made remarkable strides in machine learning, particularly excelling in diverse domains like object detection, image classification, image segmentation, facial recognition and sentiment analysis, see [8–12]. However, addressing real-world challenges remains daunting for classical machine learning approaches. QML introduces a revolutionary domain that diverges from classical methods, offering a new paradigm built on machine learning algorithms and fundamental quantum mechanics principles like superposition, quantum interference and entanglement, see [13, 14]. While classical machine learning algorithms have limitations in terms of computational complexity and time constraints, QML offers a promising alternative by utilizing quantum computing, see [15]. Among various QML algorithms, we have QCNNs, which are a family of QML that has recently become a very active field; they use a quantum circuit-based algorithm inspired by CNNs and have the advantage of efficient training and implementation on near-term quantum devices. And they have the potential for experimental realization and generalizations, see [16]. QCNN is a promising area that has many applications such as image classification, object detection, and medicine, see [17–19]. It is desired because it holds the promise of solving problems that are beyond the reach of classical systems, providing exponential speedups, improving optimization processes, and enabling new types of algorithms that could revolutionize fields like AI, healthcare, materials science, and more. While classical machine learning is still extremely powerful and widely used, QCNN has the potential to push the boundaries of what's possible and provide significant advantages in the near and long-term future.

Sentiment analysis is a key Natural Language Processing (NLP) task, but classical models like CNNs often require large datasets and deep architectures to handle complex language patterns. QCNNs offer a promising alternative by using quantum principles such as entanglement and superposition to capture subtle sentiment cues more efficiently. In addition QCNNs may generalize better in low-data settings and provide more expressive models, making them a compelling direction for future sentiment analysis as quantum technology advances. Motivated by these advantages, our main goal in this paper is to develop a new QCNN model for sentiment analysis by using different quantum encoding methods with different parametrized circuits. Encoding classical text data into quantum states is a foundational step in QCNN. Among the various encoding schemes, this work considers amplitude encoding, angle encoding, and IQP encoding.

As we can see, QCNNs offer some exciting potential advantages, but there are several limitations when applying them to text classification tasks such as: hardware limitations, quantum data representation, and training complexity.

The contributions of this study are as follows:

- Proposing a novel approach for sentiment analysis by developing a QCNN, a combination of quantum machine learning and CNN.
- Introducing three models of QCNN differentiated by the number of qubits used with utilizing both Pennylane and QISKIT frameworks, employing predefined embedding and ansatz methodologies.
- Evaluating the proposed models using key performance metrics, including accuracy, precision, and F1 score, and making comparisons between them.

The other sections of the paper are organized as follows: In Sect. 2, we outline the various related works of the model. In Sect. 3, we will explore some fundamental aspects of quantum computers. In Sect. 4, we describe our model. In Sect. 5, we test our model on two datasets and we discuss the results. Finally, Sect. 6 summarizes the paper with some potential future works.

2 Related work

Quantum machine learning has the potential to be used in various applications, including sentiment analysis. Here's in Table 1, how QML could be utilized for sentiment analysis in different works.

As mentioned above QCNN is one of QML approaches, researchers are exploring how to apply QCNNs in divers domains such as image classification, but there is no application yet of this approach in sentiment analysis. The concept of a QCNN was first introduced by [16]; it applies a CNN structure to a quantum computer. The QCNN model of this work performs the convolutional layer operation followed by the pooling layer, which are the main components of QCNN. The concept unfolds as follows:

Table 1 Some works in QML and quantum-inspired machine learning approaches for sentiment analysis

Approach	Quantum encoding	Dataset	Author
GQLM	n/a	OMD SS-tweet	Zhang et al. [20]
Quantum-inspired Interactive Networks(QIN)	n/a	MELD IEMOCAP	Zhang et al. [21]
TextTN	n/a	MR CR Subj MPQA	Zhang et al. [22]
BERT+TextTN	n/a	SST-2 SST-5	Zhang et al. [22]
QNet	Positional encoding	ColBERT	Day et al. [23]
QFNN	Angle encoding	CVTD , GSTD	Dave et al.[24]

Furthermore, n/a stands for not available

(a) The convolution circuit determines the hidden state by applying multiple qubit gates between neighboring qubits.
(b) The pooling circuit decreases the size of the quantum system by applying 2-qubit gates such as CNOT gate.
(c) Repeat the defined convolution and pooling circuits in a)–b).
(d) When the system size is sufficiently reduced, the fully connected circuit provides the classification result.

The model used in this structure is Multiscale Entanglement Renormalization Ansatz (MERA), see [25], which is a model designed for the efficient simulation of many-body quantum systems. QCNN applies MERA in reverse, exponentially reducing the size of the quantum system based on the given data.

After the appearance of QCNN, many applications of this model were created, among them the work of [17], which proposed the QCNN for image classification, it uses only two-qubit interactions throughout the entire algorithm and achieves high classification accuracy with a minimal number of parameters. This study benchmarks the performance of the parameterized QCNN across various factors using the MNIST and Fashion MNIST datasets on Pennylane. The benchmarking of quantum encoding also considers dimensionality reduction methods, which are essential for nascent quantum computers constrained by a limited number of qubits. This work evaluate the performance of the QCNN algorithm using different quantum data encoding techniques like amplitude encoding, angle encoding, qubit encoding, dense encoding, and hybrid encoding. All QCNN model achieved high classification accuracy, with the best results being approximately 94% for Fashion MNIST and about 99% for MNIST. The work includes a comparison of the classification performances of QCNNs and CNNs which reveals that QCNNs achieved excellent classification accuracy, outperforming the CNN models.

In our work, we present a novel approach to sentiment analysis tasks by using QCNN model, applying Word2Vec for word embedding and using three techniques of quantum encoding : amplitude encoding, angle encoding, and IQP encoding, also we adapt some of the circuits used for the convolution and pooling layers in the work of [17].

3 Background

This section provides an overview of the main components of quantum computing. We describe the concepts and mathematical definitions of a qubit, superposition, quantum gates, and entanglement.

3.1 Qubit

A qubit is the fundamental unit of quantum information, which can be viewed as zeros and ones in a fuzzy combination. In theory, quantum bits could be represented

by any system that behaves according to quantum principles, such as an electron or atom, see Fig. 1. Specifically, a qubit can be represented in the following ways:

- An electron orbiting a nucleus, where the $|0\rangle$ state represents the ground state, and the $|1\rangle$ state represents the excited state.
- A photon, where the $|0\rangle$ and $|1\rangle$ states correspond to its different polarization states.

So, a qubit may be defined as the complex two-dimensional Hilbert space. The qubit states $|0\rangle$ and $|1\rangle$ can be represented with vectors, as shown below. These are known as the computational basis of a two-level system:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

3.2 Superposition

A qubit can be represented as a superposition of two states at any given time, regardless of its initialization or the moment of measurement, as: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

Qubits can exist in a superposition of both basis states, $|0\rangle$ and $|1\rangle$. When a qubit is measured, it collapses to one of its eigenstates, and the measured value reflects that specific state. For example, a measurement will cause a qubit to collapse into either state $|0\rangle$ or state $|1\rangle$, with equal probability of 50%, when it is in a superposition of these two states with equal weights.

3.3 Quantum gates

Quantum gates involve the modification of one or more qubits. A binary gate is formed by an operator box that connects two quantum wires. Table 2 summarizes the most employed quantum gates.

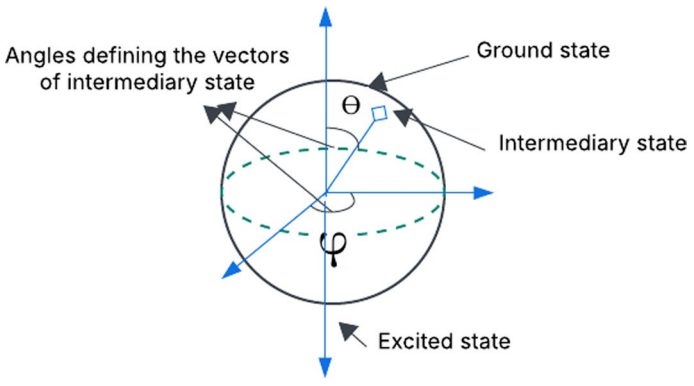
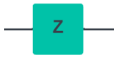
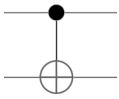


Fig. 1 Mathematical representation of qubit

Table 2 Quantum gates

	Name	Circuit schema	Notation
Single-Qubit Gates	Hadamard		$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
	Pauli-X		$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
	Pauli-Y		$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
	Pauli-Z		$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
	X-Rotation		$R_x(p) = \begin{bmatrix} \cos\left(\frac{p}{2}\right) & -i\sin\left(\frac{p}{2}\right) \\ -i\sin\left(\frac{p}{2}\right) & \cos\left(\frac{p}{2}\right) \end{bmatrix}$
	Y-Rotation		$R_y(p) = \begin{bmatrix} \cos\left(\frac{p}{2}\right) & -\sin\left(\frac{p}{2}\right) \\ \sin\left(\frac{p}{2}\right) & \cos\left(\frac{p}{2}\right) \end{bmatrix}$
	Z-Rotation		$R_z(p) = \begin{bmatrix} e^{-ip/2} & 0 \\ 0 & e^{ip/2} \end{bmatrix}$
Multi-Qubit Gate	CNOT		$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

3.4 Entanglement

Quantum entanglement is a phenomenon where quantum particles become connected in such a way that their states are dependent on each other, regardless of the distance between them. An entangled pair is a system of two quantum particles whose quantum states are strongly correlated in such a way that measuring one instantly determines the state of the other, no matter how far apart they are.

The Bell states are quantum states involving two qubits that represent fundamental forms of quantum entanglement, forming an orthonormal basis:

$$\left|\psi^{00}\right\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \qquad \left|\psi^{01}\right\rangle := \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle),$$
$$\left|\psi^{10}\right\rangle := \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \qquad \left|\psi^{11}\right\rangle := \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$

4 Methods

In this section, we applied QCNNs in a movie reviews dataset to predict the positive and negative sentences. The overall proposed architecture of the classifier QCNN can be seen in Fig. 2.

Here is a detailed explanation of each component of the model's architecture.

4.1 Preprocessing

After the collection of data, we apply this process which consists of two main steps: preprocessing and vector generation. The first step involves converting the data into a format that is easily understandable by the user. Sentiment analysis datasets are vast and often contain incomplete and inconsistent informations due to their diverse sources. Therefore, the preprocessing consists of cleaning the data, by splitting the text into tokens, removing punctuations, stop words, and single characters, and converting the text to lowercase, the cleaned data are then ready for vector generation. The second step is vector generation or rather Word Embedding which is a method that attempts to transform a word to a vector. Previous research has commonly utilized bag-of-words [26] and TF-IDF [27] methods for word embedding, but due to their high dimensionality, current research prefers the use of Word2Vec and GloVe [28] models.

In our work, we use the Word2Vec embedding which trains each word in the corpus and converts it into a d-dimensional vector (here $d = 300$).

Word2Vec is a neural network-based approach used to compute continuous distributed representations of words. The model efficiently implements the Continuous Bag of Words (CBOW) and Skip-gram architectures to compute vector representations of words. Both architectures are simple neural network models with one hidden layer. The method first generates a vocabulary from the input words, and the word vectors are learned through backpropagation and stochastic gradient descent, see [29].

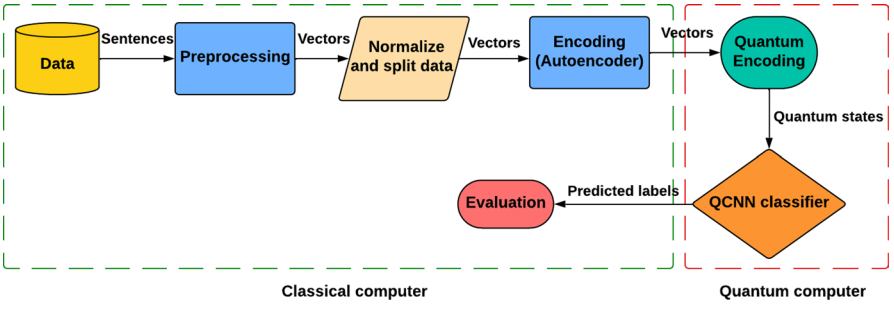


Fig. 2 General architecture of the proposed model

4.2 Standardize and split data

Standardizing a dataset or centering the data involves rescaling the distribution of values so that the mean is 0 and the standard deviation is 1. This process involves subtracting the mean value from each observation. Standardization can be useful and even necessary when the input values have differing scales. It assumes that the data follow a Gaussian distribution with a well-defined mean and standard deviation. After the process of standardizing, the data are divided into two categories: training sets and testing sets. The first category represents 33% of total reviews and the second represents the rest.

4.3 Encoding (autoencoder)

In our work, it is necessary to reduce the dimensionality of word vectors to make them compatibles to be classified with QCNN. There are many methods of dimensionality reduction, among them the autoencoders.

An autoencoder consists of two main components: the encoder and the decoder. The encoder processes the input data and converts it into a lower-dimensional representation known as the latent space. The decoder then takes this latent representation and attempts to reconstruct the original input as accurately as possible.

So, the encoder component of autoencoder reduces dimensionality by encapsulating the most significant features and patterns in the data while eliminating noise and redundancy. In our work, we reduce the dimensionality of data vectors from dimension = 300, which is the dimension of the word vector to the dimension of the QCNN classifier (4, 8, or 16); by using a simple autoencoder with one hidden layer, the data in the latent space are then fed to quantum encoding, as illustrated in Fig. 3.

4.4 Quantum encoding

The vectors generated from dimensionality reduction are encoded and transformed into quantum states with a quantum encoding method; in this work, we evaluate the performance of the QCNN algorithm using different quantum data encoding methods. A detailed explanation of these techniques is provided in this section.

4.4.1 Angle encoding

It is the most widely used approach due to its simplicity; this method utilizes the relationship where the probability of measuring a specific state is influenced by its phase. Quantum states are represented by complex numbers, where the amplitudes indicate the probabilities of different measurement outcomes. The phase of these amplitudes, depicted as angles, holds significance in quantum algorithms.

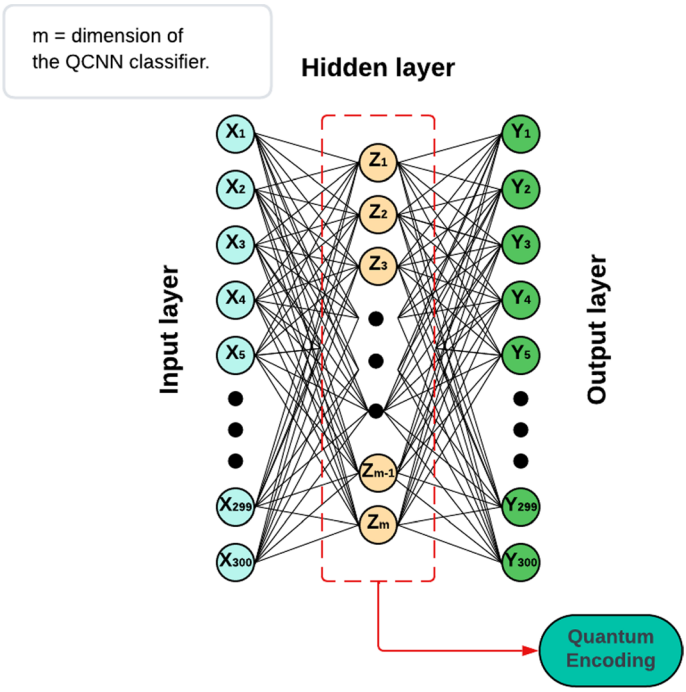


Fig. 3 Autoencoder architecture of the model

Angle encoding facilitates the integration of classical data into quantum states, empowering quantum computers to manipulate classical information. Let us consider a single-qubit state $|\psi\rangle$ and an angle θ. The angle embedding operation can be written as:

$$|\psi(\theta)\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle + \cos\left(\frac{\theta}{2}\right)|0\rangle - e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle.$$

This approach is particularly well-suited for embedding word representations or contextual features, as it maintains the relative relationships within the vector space, a characteristic that is especially important for capturing sentiment, which frequently emerges from nuanced contextual shifts.

4.4.2 Amplitude encoding

This method consists to map the normalized input data to the probability amplitudes of a quantum state. The input data $x = (x_1, x_2, \ldots, x_N)^T$ of dimension $N = 2^n$ are represented as the amplitude of an n-qubit quantum state $|\phi(x)\rangle$.

$$U_\phi(x) : x \in \mathbb{R} \rightarrow |\phi(x)\rangle = \frac{1}{||x||} \sum_{i=1}^{N} x_i |i\rangle,$$

where $|i\rangle$ is the ith computational basis state. Clearly, with amplitude encoding, a quantum computer can represent an exponentially large amount of classical data. This offers a significant advantage in QCNN algorithms. As the number of parameters that need to be optimized scales as $O(\log(n))$, amplitude encoding reduces the number of parameters in a doubly exponential manner relative to the size (or dimension) of the classical data.

We use this method because it is widely used and effective for embedding dense numerical features such as : sentiment polarity scores or word vector magnitudes, but it may lose fine-grained emotional features due to amplitude squashing.

4.4.3 IQP encoding

The IQP mapping technique draws inspiration from physics and maps data points into a second-order Hamiltonian. We assume that $x \in (0, 2\pi]^n$ and note the approach suggests a quantum advantage. Taking the function f in the equation below and defining the function $\tilde{f} = 2\pi \cdot f$ that maps $\tilde{f} : \mathbb{R}^{n-1} \rightarrow (0, 2\pi]^n$ does not lose information since f is a homeomorphism. Denoting Z_i as the Pauli Z gate acting on the ith qubit the authors define the specific unitary operator

$$U_Z(x) = \exp\left(\sum_{i=1}^{n} x_i Z_i + \sum_{i=1}^{n} \sum_{j=1}^{n} (\pi - x_i)(\pi - x_j) Z_i Z_j \right).$$

One may note the coefficients in the quadratic terms are centered around 0 with standard deviation of 1, and the original data may be mapped accordingly into a different form. Denoting H as the Hadamard gate, the IQP encoding is defined as

$$D_i \rightarrow U_Z(D_i) H^{\oplus n} U_Z(D_i) H^{\oplus n} |0 \ldots 0\rangle.$$

Particularly, IQP encoding is interesting for encoding nonlinear and entangled relationships in text, offering a potentially richer representation of complex sentiment expressions, like irony or sarcasm.

4.4.4 Comparative analysis

In this subsection, we make a comparison between quantum encoding methods (refer to [30–36]).

From Table 3, it is evident that while amplitude encoding offers simplicity and efficiency, it may not sufficiently preserve the nuanced emotional characteristics inherent in natural language. Angle encoding strikes a balance between interpretability and sentiment retention, especially when dealing with vectorized representations like word embeddings. IQP encoding, despite its implementation complexity, shows promise in modeling deeper linguistic patterns and subtle sentiment shifts,

such as sarcasm or indirect emotional cues, due to its ability to generate complex, entangled quantum states.

4.5 Quantum convolutional neural network classification

In this section, we represent three QCNN classifiers differentiated by their size of input and the number of layers, as shown in Figs. 4, 5, and 6, every QCNN classifier starts with the word embedding which is encoded onto a quantum computer by a quantum encoding method, then, the convolution and pooling layers are applied, respectively, and repeatedly on the quantum state generated until the system size is sufficiently small, and finally, the measurement is applied to every qubit to take the quantum state back to a classical computer.

The QCNN consists of two main parts, the convolution and the pooling:

1. Convolution. Every QCNN model requires selecting an ansatz, which is a two-qubit unitary operation, knowing that our model allows for flexibility in applying any ansatz for each convolutional filter. However, we use the same ansatz across all convolutional filters and pooling operations. While employing different ansatz for each filter presents an exciting opportunity for further enhancements, it also increases the number of parameters that require optimization. In this work, we use 3 different unitaries for convolution; all unitaries are characterized by balanced significance in both expressibility and entangling capability as shown in the work of [17]. Figure 7 shows the architecture of each unitary.

2. Pooling. The pooling layer applies parameterized pooling unitary to two qubits and subsequently discards one of the qubits, thereby reducing the two-qubit states to a one-qubit state. There are a variety of circuits of the pooling unitary, but in our study, we opt for a simplified version of this circuit from the work of [17]. This unitary is shown in Fig. 8.

The pooling layer applies two controlled rotations $R_z(p_1)$ and $R_z(p_2)$, respectively, each activated when the control qubit is 1 (filled circle) or 0 (open circle). The first qubit is traced out after the gate operations in order to reduce the dimension.

5 Results and discussion

In our experiments, we focus on movie reviews that have short sentence input, 2/3 from dataset are used for training, while others are used for testing, all input sentences will be preprocessed. The experiments are performed with two datasets, namely *real dataset* and *artificial dataset*. Our benchmark is centered on binary classification; for generating word vectors, a pre-trained Google News dataset from Word2Vec (which is publicly available) has been used. The description of the experimented datasets is as follows:

1. Reel dataset. We use the Sentence polarity dataset movie review which has retrieved from imdb.com. The size of the longest sentence in this dataset is 59. We use 7137 sentences as training sets and 3516 sentences as testing sets.

Table 3 Comparison between quantum encoding methods

Criterion	Amplitude encoding	Angle encoding	IQP Encoding
Encoding Efficiency	High (compact), but normalization required	Moderate	Low
Information Loss	Moderate (amplitude compression)	Low	Low to Moderate
Sentiment Feature Retention	Low to Moderate	High (captures relative context well)	High
Complexity of Implementation	Low to Moderate	Moderate	High
Suitability for Emotionally Rich Text	Limited	Good	Very good

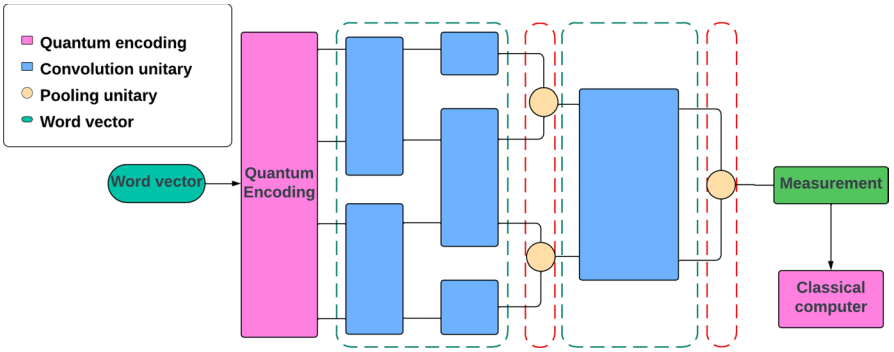


Fig. 4 The architecture of QCNN with input of 4 qubits. This quantum circuit contains: quantum data encoding which is represented by pink rectangle, convolutional unitaries which are represented by blue rounded rectangles, and pooling unitaries which are represented by orange circles. The convolutional and pooling operations in this circuit are implemented using parameterized quantum gates. The circuit consists of two layers, each allowing the application of multiple convolutional unitaries. Within a layer, each convolutional unitary applies the same two-qubit ansatz to pairs of nearest-neighbor qubits in a translationally invariant manner. Pooling operations in each layer also use the same ansatz. In this architecture, pooling is realized as a controlled unitary operation that activates when the control qubit is in the state 1

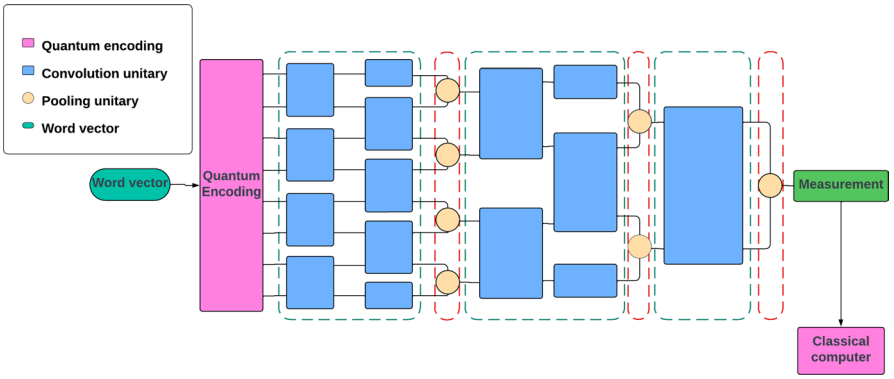


Fig. 5 The architecture of QCNN with input of 8 qubits. The circuit consists of three layers, and its function is similar to Fig. 4

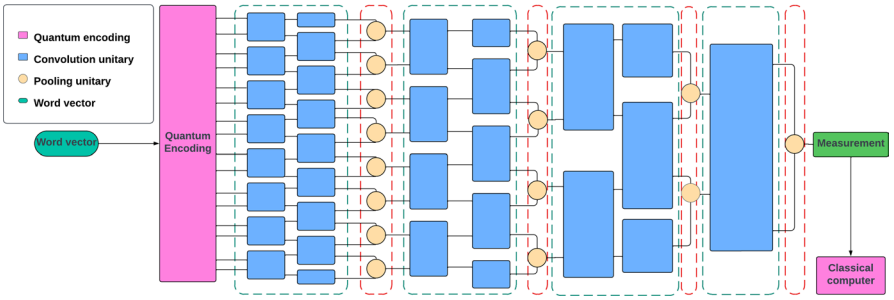
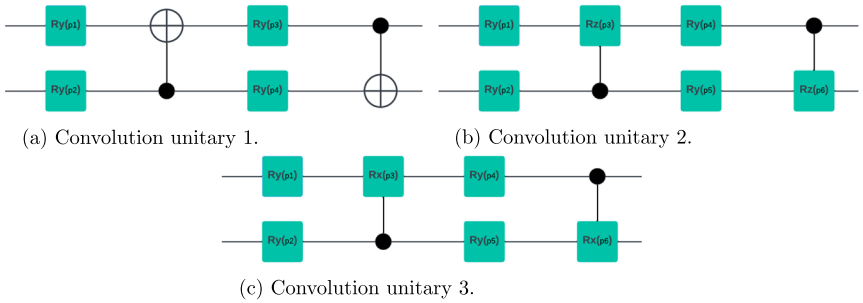


Fig. 6 The architecture of QCNN with input of 16 qubits. This quantum circuit contains four layers



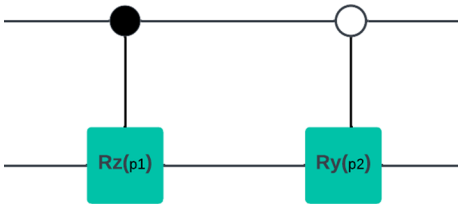
(a) Convolution unitary 1.

(b) Convolution unitary 2.

(c) Convolution unitary 3.

Fig. 7 Convolution unitaries

Fig. 8 Pooling unitary



2. Artificial dataset. We use a dataset from the work of [37]. The size of the longest sentence in this dataset is 7. In this research, we use 134 sentences as training sets and 66 sentences as testing sets.

5.1 Experimental configuration

We implemented QCNN with TensorFlow, QISKIT, and Pennylane. The hardware specifications used at the time of experiments are shown in Table 4.

5.2 QCNN results in artificial dataset

Table 5 shows the best classification accuracy, precision, and F1 score obtained from artificial dataset with random initialization of parameters and different QCNN classifiers. The number of random initializations is chosen to be the same as that used by [17]. The highest accuracy in each column is indicated in bold font.

Table 4 Exeuction hardware

Item	Detail
OS	Microsoft Windows 11, 64 bit
CPU	AMD Ryzen 74700U with Radeon Graphics
RAM	16 GO
GPU	Not available
TPU	Not available
Application	Python 3.12.7

5.3 QCNN results in real dataset

Here, in Table 6, the best classification accuracy, precision, and F1 score obtained from real dataset with random initialization of parameters and different QCNN classifiers. The highest accuracy in each column is indicated in bold font.

5.4 Comparison with CNN

From Table 7, we observe that QCNN models outperform their corresponding CNN models for both datasets.

5.5 Analysis of experimental results

The results obtained by training QCNN of 4 qubits in artificial dataset show that the ansatz 3 achieved the best accuracy 90,9% with angle encoding. The accuracies obtained by training QCNN of 8 qubits in artificial dataset show that the performance of the classifier has been ameliorated compared with QCNN of 4 qubits; the ansatz 3 achieved the best accuracy 92,4% with angle encoding. So, the angle encoding was more performant than the other quantum encoding techniques if we use 4 or 8 qubits, while amplitude encoding gives close and weak accuracies. As shown in Table 5, the QCNN of 16 qubits has been achieved the best performance compared with the other QCNN models; the best quantum encoding technique for this model is angle encoding too. So, with the increase of the number of qubits, the accuracy improves, and generally the ansatz 3 gives better results. Noting that, we cannot applied QCNN of 16 qubits with amplitude encoding method because $2^{16} > 300$ (the size of a word vector).

The results obtained by training QCNN in real dataset show that the accuracies are close to each other for all encoding methods and the different ansatz. However, in QCNN of 4 and 8 qubits, the use of ansatz 1 with the angle encoding method achieves the best accuracy $52,0\%$ and $53,3\%$, respectively, while, in QCNN of 16 qubits, the use of the ansatz 3 with IQP encoding gives the best accuracy of $51,9\%$. In general, the angle encoding was more performant than the other quantum encoding methods. As shown in Table 6, the results obtained are

Table 5 Classification accuracy, precision, and F1 score obtained with QCNN in artificial dataset

Number of qubits	Ansatz	Number of params	Quantum encoding	Accuracy (%)	Precision (%)	F1 score (%)	Runtime (s)
4	1	12	Angle	83.3	84.6	83.3	0.95
	2	16	Angle	89.4	89.8	89.3	1.00
	3	16	Angle	**90.9**	**90.9**	**90.9**	1.29
	1	12	Amplitude	63.6	63.7	62.8	0.89
	2	16	Amplitude	63.6	63.5	63.3	0.97
	3	16	Amplitude	**69.7**	**69.6**	**69.4**	1.23
	1	12	IQP	78.8	82.7	78.5	1.00
	2	16	IQP	72.7	77.8	70.5	1.07
	3	16	IQP	**83.3**	**85.1**	**82.9**	1.26
8	1	18	Angle	92.4	92.5	92.4	2.44
	2	24	Angle	81.8	84.6	81.7	2.53
	3	24	Angle	**92.4**	**93.0**	**92.3**	2.88
	1	18	Amplitude	63.6	63.5	63.5	2.10
	2	24	Amplitude	65.2	65.3	65.1	2.30
	3	24	Amplitude	**66.7**	**66.7**	**66.6**	2.83
	1	18	IQP	81.8	82.7	81.8	2.72
	2	24	IQP	84.8	87.8	84.7	2.74
	3	24	IQP	**86.4**	**88.7**	**86.3**	3.69
16	1	24	Angle	87.9	88.8	87.9	19.97
	2	32	Angle	**90.9**	**90.9**	**90.9**	23.66
	3	32	Angle	89.4	89.4	89.3	26.44
	1	24	Amplitude	/	/	/	/
	2	32	Amplitude	/	/	/	/
	3	32	Amplitude	/	/	/	/
	1	24	IQP	71.2	75.0	70.7	33.29
	2	32	IQP	**90.3**	80.5	80.1	35.36
	3	32	IQP	**90.9**	**91.0**	**90.8**	37.22

converging, so the change in encoding methods or the QCNN model or even the ansatz used does not affect much the obtained performances.

As shown in Tables 5 and 6, the accuracies obtained in artificial dataset are more better than those obtained in real dataset, this is an expected result due to the size of the sentence (longest sentence in real dataset is 59, while it is just 7 in artificial data set), more the size is short, more the classifier is better.

As shown in Fig. 9, the number of qubits does not affect much the accuracy; this is an expected result due to the capacity of QCNN to generalize from less deep architecture.

Tables 5 and 6 show that the runtime became more important with the increase of qubits which affect the number of parameters. In addition, the IQP encoding was

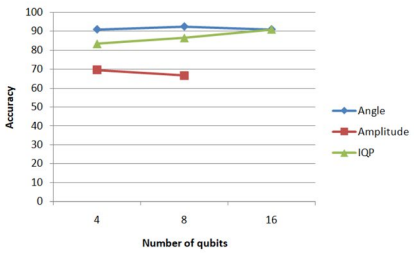
Table 6 Classification accuracy, precision, and F1 score obtained with QCNN in real dataset

Number of qubits	Ansatz	Number of params	Quantum encoding	Accuracy (%)	Precision (%)	F1 score (%)	Runtime (s)
4	1	12	Angle	**52.0**	**52.0**	**52.0**	478.36
	2	16	Angle	51.8	51.8	51.7	555.13
	3	16	Angle	51.8	51.8	51.8	667.05
	1	12	Amplitude	51.2	51.2	51.2	481.84
	2	16	Amplitude	**51.3**	**51.3**	**51.3**	537.72
	3	16	Amplitude	50.6	50.6	50.6	600.99
	1	12	IQP	51.4	51.5	50.0	544.71
	2	16	IQP	**51.5**	**51.5**	**51.1**	1329.87
	3	16	IQP	51.1	51.5	48.2	1653.99
8	1	18	Angle	**53.3**	**53.5**	**51.2**	737.31
	2	24	Angle	53.2	53.4	50.2	3484.80
	3	24	Angle	53.1	53.0	52.1	3318.79
	1	18	Amplitude	**51.7**	**51.7**	**51.6**	681.24
	2	24	Amplitude	51.3	51.3	51.1	1346.31
	3	24	Amplitude	51.1	51.1	51.1	2104..01
	1	18	IQP	**51.7**	**51.7**	**51.5**	972.95
	2	24	IQP	50.8	51.4	46.3	3578.74
	3	24	IQP	51.0	51.0	50.7	3867.56
16	1	24	Angle	51.8	51.8	50.9	11469.96
	2	32	Angle	**51.8**	**51.8**	**51.8**	12050.21
	3	32	Angle	51.0	43.2	50.4	12150.64
	1	24	Amplitude	/	/	/	/
	2	32	Amplitude	/	/	/	/
	3	32	Amplitude	/	/	/	/
	1	24	IQP	51.2	51.2	50.7	12030.96
	2	32	IQP	50.6	51.5	43.4	13430.96
	3	32	IQP	**51.9**	**52.0**	**51.7**	13887.96

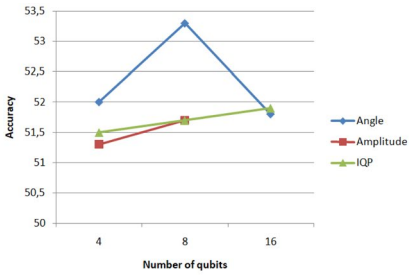
Table 7 Comparison of QCNN and CNN in artificial and real datasets

Model	Dataset	Epoch	Accuracy (%)
QCNN	Artificial	200	**92.4**
CNN	Artificial	200	81.8
QCNN	Reel	200	**53.3**
CNN	Reel	200	50.5

slower than the other quantum encoding method. And the amplitude encoding is faster but less performant.



(a) Classification accuracy according to the number of qubits in the Artificial dataset.



(b) Classification accuracy according to the number of qubits in the real dataset.

Fig. 9 Comparison of best accuracy according to the number of qubits and quantum encoding method

6 Conclusion

This paper explores the potential of quantum QCNN for sentiment analysis, presenting the first implementation of this algorithm with varying the number of qubits and the quantum encoding method in the field of NLP. Despite the promising results, several challenges persist particularly regarding scalability, computational complexity, and hyperparameter tuning.

Our experimental results indicate that QCNN represents a compelling QML approach to sentiment analysis. Notably, the model achieved higher accuracy on an artificially generated dataset compared to a real-world dataset, suggesting that performance is significantly influenced by both dataset characteristics and sentence length. In addition, despite the slowness of QCNN, it has the ability to generalize from a less deep architecture. For future research, we plan to extend this work by evaluating QCNNs on datasets with diverse sentence lengths, incorporating more advanced convolution and pooling unitaries with layer-wise variations, and exploring multi-class classification tasks in NLP. These directions hold considerable promise for advancing QML in natural language understanding also using QCNN models for multi-class classification in NLP which remains an exciting way for future research.

Author contributions All authors participate in the manuscript.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical Statement This research was done according to ethical standards.

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